A Geospatial Cloud

for scalable, reproducible science

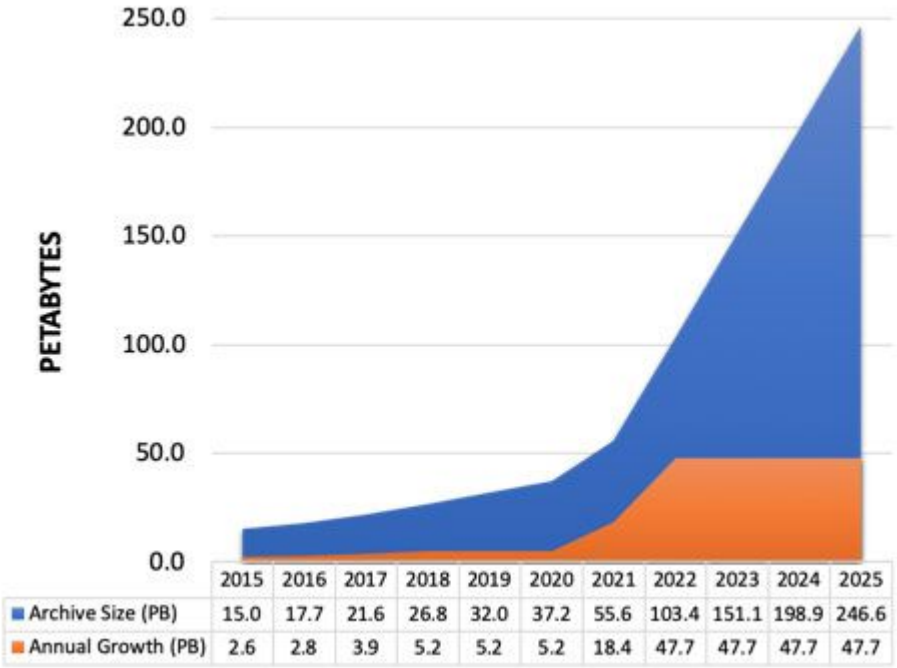
SpatioTemporal Asset Catalogs
+
Cloud-Optimized GeoTiffs

Landsat Science Team Winter Meeting 2020

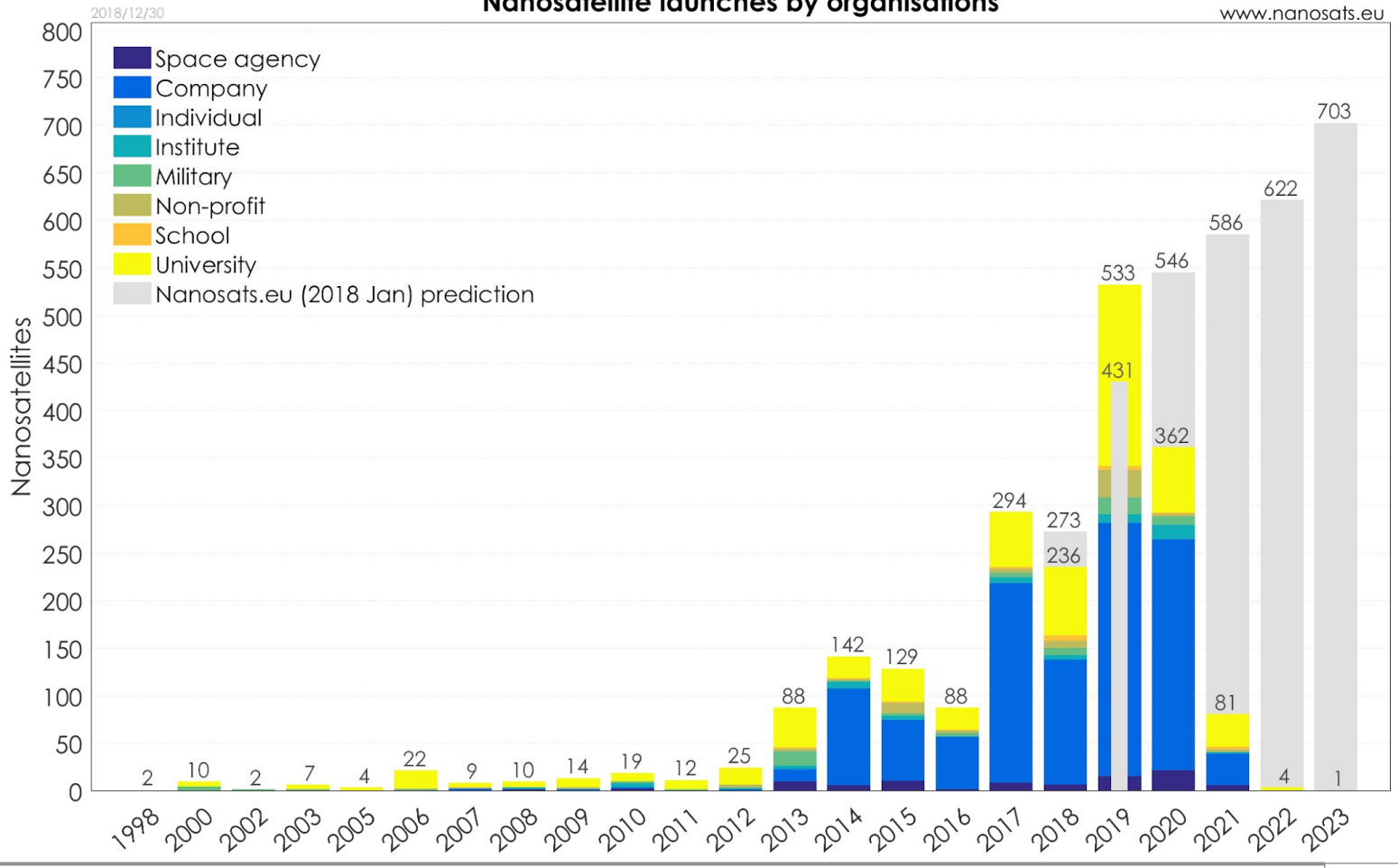
Matthew Hanson @GeoSkeptic

Element 84

NASA EOSDIS Archive



	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Archive Size (PB)	15.0	17.7	21.6	26.8	32.0	37.2	55.6	103.4	151.1	198.9	246.6
Annual Growth (PB)	2.6	2.8	3.9	5.2	5.2	5.2	18.4	47.7	47.7	47.7	47.7


Nanosatellite launches by organisations
2018/12/30
www.nanosats.eu
Nanosatellites
Space agency
Company
Individual
Institute
Military
Non-profit
School
University
Nanosats.eu (2018 Jan) prediction
800
750
700
650
600
550
500
450
400
350
300
250
200
150
100
50
0
2
10
2
7
4
22
9
10
14
19
12
25
88
142
129
88
294
273
236
533
431
546
362
586
81
622
4
703
1
1998
2000
2002
2003
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023

```
LC08_L1GT_168005_20190828_20190828_01_RT.tif
LC08_L1GT_168005_20190828_20190828_01_RT_B1.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_B3.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_c10_f1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_cm.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_class_c50_i9.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_tmp_test5.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_ver7.6.tif
LC08_L1GT_168005_20190828_20190828_01_final_class2.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_class.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_final.tif
LC08_L1GT_168005_20190828_20190828_01_test.tif
LC08_L1GT_168005_20190828_20190828_test1.tif
LC08_L1GT_168005_20190828_20190828_test_test.tif
```

New Approaches are Needed

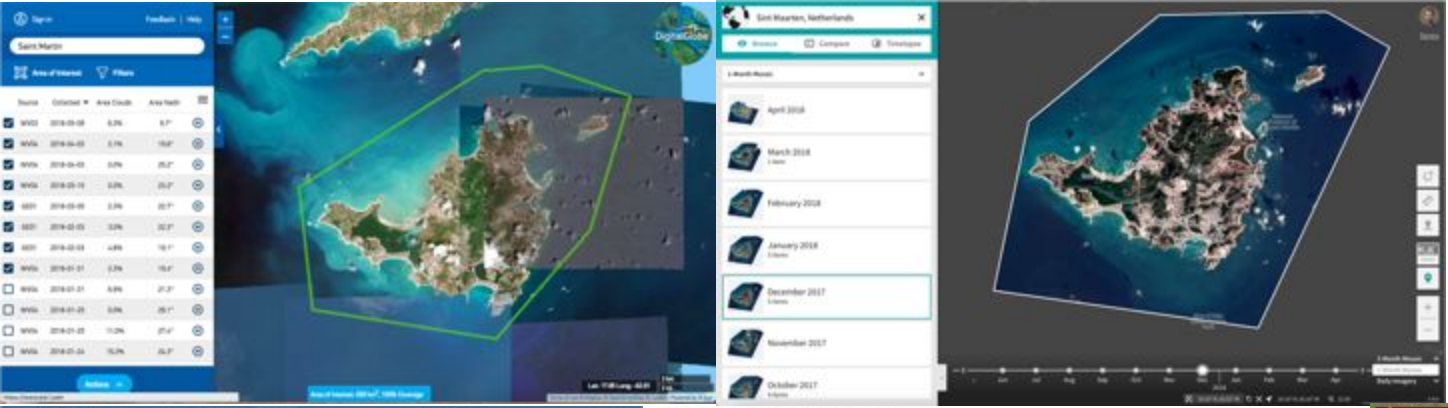
- If your (meta)data cannot be crawled and indexed, your data might as well not exist
- If data provenance cannot be determined, your research cannot be reproduced
- If you need to download files, you will not be able to scale processing

SpatioTemporal Asset Catalogs

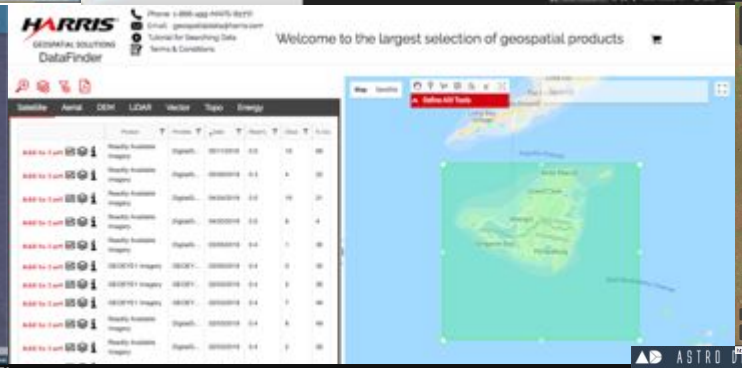


A specification to improve interoperability of geospatial data

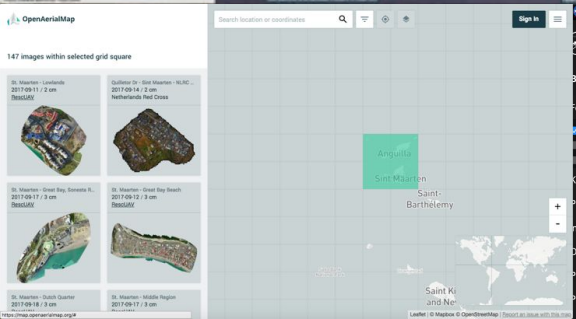
Why?


Saint Martin
Sint Maarten, Netherlands
April 2018
March 2018
February 2018
January 2018
December 2017
November 2017
October 2017

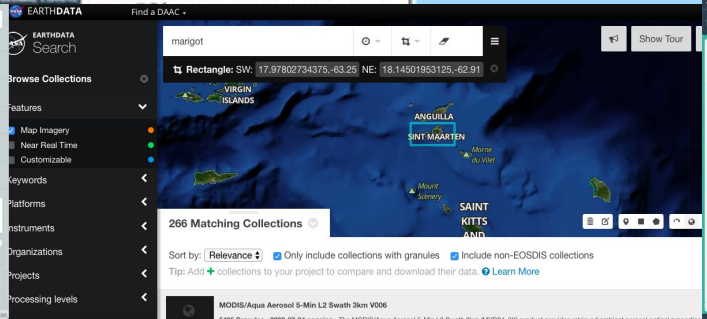

SENTINEL Hub Playground

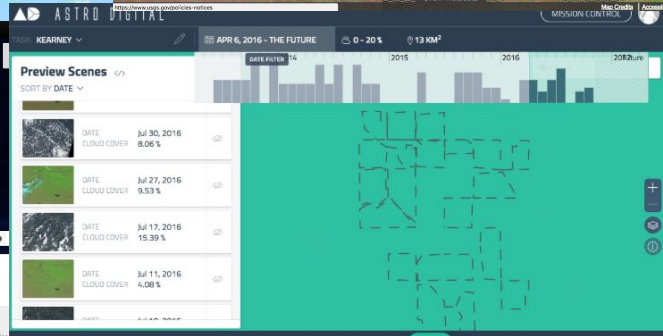

HARRIS GEOSPATIAL SOLUTIONS
DataFinder
Welcome to the largest selection of geospatial products

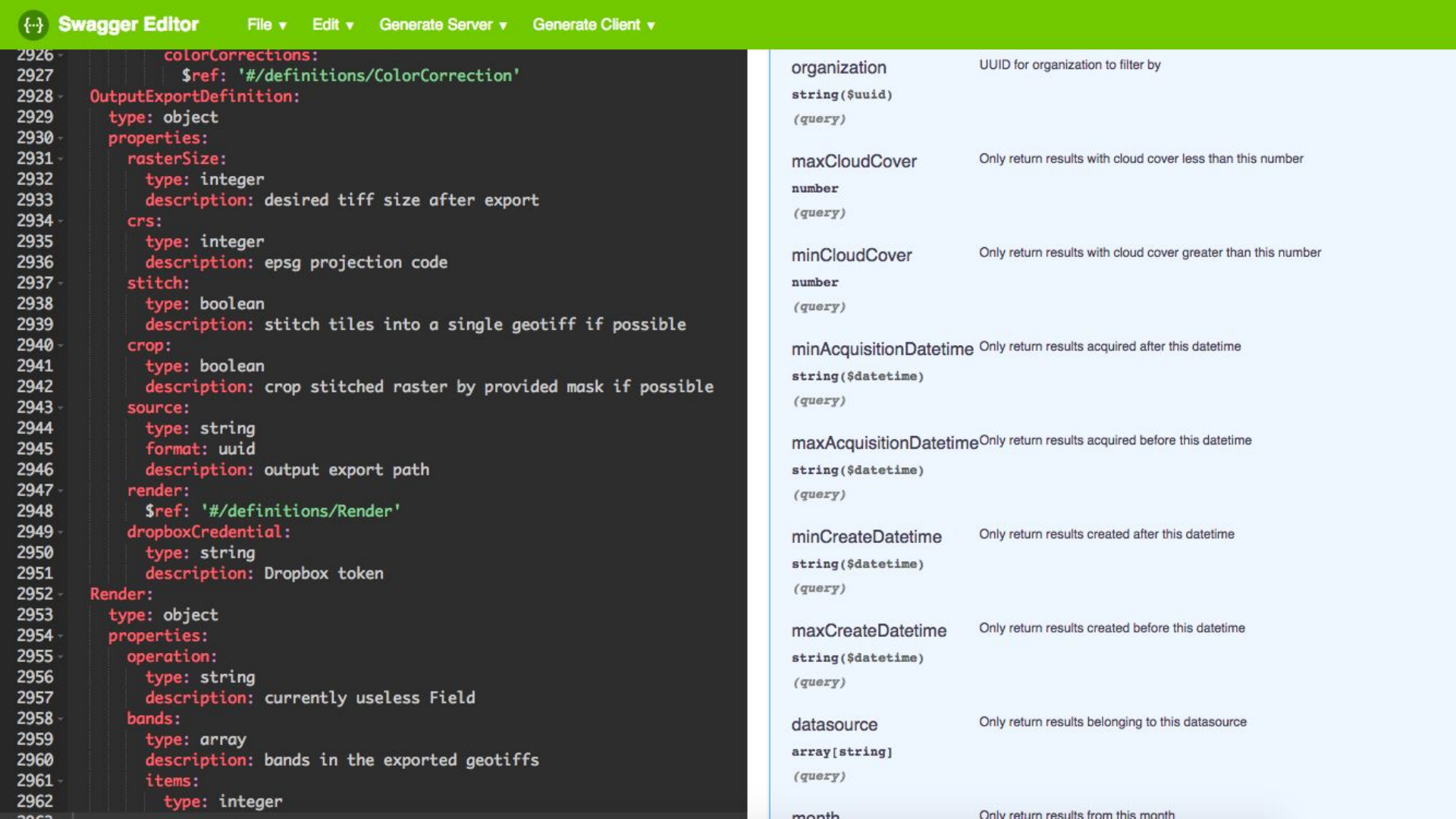

USGS LandsatLook
Show Metadata


OpenAerialMap
147 images within selected grid square

Anguilla
Sint Maarten
Saint-Barthelemy


EARTHDATA
Search
marigot
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
Show Tour
Browse Collections
Features
Map Imagery
Near Real Time
Customizable
Keywords
Platforms
Instruments
Organizations
Projects
Processing levels
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006


ASTRO DIGITAL
MISSION CONTROL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
Jul 30, 2016
8.06 %
Jul 27, 2016
9.93 %
Jul 17, 2016
15.39 %
Jul 11, 2016
4.08 %


Swagger Editor
File
Edit
Generate Server
Generate Client
colorCorrections:
$ref: '#/definitions/ColorCorrection'
OutputExportDefinition:
type: object
properties:
rasterSize:
type: integer
description: desired tiff size after export
crs:
type: integer
description: epsg projection code
stitch:
type: boolean
description: stitch tiles into a single geotiff if possible
crop:
type: boolean
description: crop stitched raster by provided mask if possible
source:
type: string
format: uuid
description: output export path
render:
$ref: '#/definitions/Render'
dropboxCredential:
type: string
description: Dropbox token
Render:
type: object
properties:
operation:
type: string
description: currently useless Field
bands:
type: array
description: bands in the exported geotiffs
items:
type: integer
organization
UUID for organization to filter by
string($uuid)
(query)
maxCloudCover
Only return results with cloud cover less than this number
number
(query)
minCloudCover
Only return results with cloud cover greater than this number
number
(query)
minAcquisitionDatetime
Only return results acquired after this datetime
string($datetime)
(query)
maxAcquisitionDatetime
Only return results acquired before this datetime
string($datetime)
(query)
minCreateDatetime
Only return results created after this datetime
string($datetime)
(query)
maxCreateDatetime
Only return results created before this datetime
string($datetime)
(query)
datasource
Only return results belonging to this datasource
array[string]
(query)

What is STAC?

SpatioTemporal Asset Catalog

- Defined metadata for geospatial catalogs and assets
  - Focus on search and discovery
  - Simple and extensible
- Linked metadata
  - Crawlable so can be indexed
  - Maintain data provenance
- Dynamic API
  - Searchable on arbitrary fields
  - OGC Features API with STAC specific extensions

What is it not?

SpatioTemporal Asset Catalog

- Full-fledged metadata standard
- Replacement for data provider's internal metadata
- Single source of truth
- Replacement for ISO 19115, OGC CSW, …

The specification

- Catalogs
  - There's always a root catalog
  - Contains other catalogs and collections
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Ultimately contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when and where)

https://earth-search.aws.element84.com/stac

+ - View source


```
{
   stac_version: "0.6.2",
   id: "earth-search",
   title: "Earth on AWS STAC API",
   description: "A STAC API powered by sat-api for Earth on AWS datasets by Element 84",
 - links: [
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-awfi"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan10m"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-mux"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/sentinel-2-l1c"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/landsat-8-l1"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan5m"
       },
     - {
           rel: "self",
           href: "https://earth-search.aws.element84.com/stac"
       }
   ]
}
```


links

Links

Field Name	Type	Description
href	string	**REQUIRED.** The actual link in the format of an URL. Relative and absolute links are both allowed.
rel	string	**REQUIRED.** Relationship between the current document and the linked document. See chapter "Relation types" for more information.
type	string	Media type of the referenced entity.
title	string	A human readable title to be used in rendered displays of the link.

Link Relation Types

Type	Description
self	STRONGLY RECOMMENDED. *Absolute* URL to the location that the catalog file can be found online, if available. This is particularly useful when in a download package that includes metadata, so that the downstream user can know where the data has come from.
root	STRONGLY RECOMMENDED. URL to the root STAC Catalog or Collection. Catalogs should include a link to their root, even if it's the root and points to itself.
parent	URL to the parent STAC Catalog or Collection. Non-root catalogs should include a link to their parent.
child	URL to a child STAC Catalog or Collection.
item	URL to a STAC Item.

Other Links: **about, derived-from, previous-version (any IANA rel type)**

Item Fields

Field Name	Type	Description
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. As most geospatial assets are already defined by some identification scheme by the data provider it is recommended to simply use that ID. Data providers are advised to include sufficient information to make their IDs globally unique, including things like unique satellite IDs.
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
geometry	GeoJSON Geometry Object	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED.** Bounding Box of the asset represented by this item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present. This field provides an easy way for a user to search for any Items that belong in a specified Collection.

Item Properties

Field Name	Type	Description
datetime	string	**REQUIRED.** The searchable date and time of the assets, in UTC. It is formatted according to RFC 3339, section 5.6.
title	string	A human readable title describing the item.
start_datetime	string	The first or start date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.
end_datetime	string	The last or end date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.

Field Name	Type	Description
platform	string	Unique name of the specific platform to which the instrument is attached.
instruments	[string]	Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
constellation	string	Name of the constellation to which the platform belongs.
mission	string	Name of the mission for which data is collected.

Item Assets

Field Name	Type	Description
href	string	**REQUIRED.** Link to the asset object. Relative and absolute links are both allowed.
title	string	The displayed title for clients and users.
description	string	A description of the Asset providing additional details, such as how it was processed or created.
type	string	Media type of the asset.
roles	[string]	The semantic role of the asset, similar to the use of `rel` in links.

Content Extensions

Extension Name (Prefix)	Scope	Description	Maturity
Asset Definition (-)	Collection	Provides a way to specify details about what assets may be found in Items belonging to a collection.	*Proposal*
Checksum (`checksum`)	Item, Catalog, Collection	Provides a way to specify file checksums for assets and links in Items, Catalogs and Collections.	*Proposal*
Commons (-)	Item, Collection	Provides a way to specify data fields in a collection that are common across the STAC Items in that collection, so that each does not need to repeat all the same information.	*Proposal*
Data Cube (`cube`)	Item, Collection	Data Cube related metadata, especially to describe their dimensions.	*Proposal*
EO (`eo`)	Item	Covers electro-optical data that represents a snapshot of the earth for a single date and time. It could consist of multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands. The extension provides common fields like bands, cloud cover, gsd and more.	*Pilot*
Label (`label`)	Item	Items that relate labeled AOIs with source imagery	*Proposal*
Point Cloud (`pc`)	Item	Provides a way to describe point cloud datasets. The point clouds can come from either active or passive sensors, and data is frequently acquired using tools such as LiDAR or coincidence-matched imagery.	*Proposal*
Projection (`proj`)	Item	Provides a way to describe items whose assets are in a geospatial projection.	*Proposal*
SAR (`sar`)	Item	Covers synthetic-aperture radar data that represents a snapshot of the earth for a single date and time.	*Proposal*
Satellite (`sat`)	Item	Satellite related metadata for data collected from satellites.	*Proposal*
Scientific (`sci`)	Item, Collection	Scientific metadata is considered to be data that indicate from which publication data originates and how the data itself should be cited or referenced.	*Proposal*
Single File STAC (-)	ItemCollection	An extension to provide a set of Collections and Items as a single file catalog.	*Proposal*
Versioning Indicators (-)	Item, Collection	Provides fields and link relation types to provide a version and indicate deprecation.	*Proposal*

EO Extension

Field Name	Type	Description
eo:gsd	number	**REQUIRED.** Ground Sample Distance at the sensor.
eo:bands	[Band Object]	**REQUIRED.** This is a list of the available bands where each item is a Band Object.
eo:cloud_cover	number	Estimate of cloud cover as a percentage (0-100) of the entire scene. If not available the field should not be provided.

EO Band Object

Field Name	Type	Description
name	string	The name of the band (e.g., "B01", "B02", "B1", "B5", "QA").
common_name	string	The name commonly used to refer to the band to make it easier to search for bands across instruments. See the list of accepted common names.
description	string	Description to fully explain the band. CommonMark 0.29 syntax MAY be used for rich text representation.
center_wavelength	number	The center wavelength of the band, in micrometers (µm).
full_width_half_max	number	Full width at half maximum (FWHM). The width of the band, as measured at half the maximum transmission, in micrometers (µm).

EO Band Common Names

Common Name	Band Range (µm)	Landsat 5/7	Landsat 8	Sentinel 2	MODIS
coastal	0.40 - 0.45		1	1	
blue	0.45 - 0.50	1	2	2	3
green	0.50 - 0.60	2	3	3	4
red	0.60 - 0.70	3	4	4	1
yellow	0.58 - 0.62				
pan	0.50 - 0.70	8 (*L7 only*)	8		
rededge	0.70 - 0.75				
nir	0.75 - 1.00	4		8	2
nir08	0.75 - 0.90		5	8a	
nir09	0.85 - 1.05			9	
cirrus	1.35 - 1.40		9	10	26
swir16	1.55 - 1.75	5	6	11	6
swir22	2.10 - 2.30	7	7	12	7
lwir	10.5 - 12.5	6			
lwir11	10.5 - 11.5		10		31
lwir12	11.5 - 12.5		11		32

Linking Assets to Bands

```
- eo:bands: [
    - {
        name: "B1",
        common_name: "coastal",
        gsd: 30,
        center_wavelength: 0.44,
        full_width_half_max: 0.02
      },
    - {
        name: "B2",
        common_name: "blue",
        gsd: 30,
        center_wavelength: 0.48,
        full_width_half_max: 0.06
      },
    - {
        name: "B3",
        common_name: "green",
        gsd: 30,
        center_wavelength: 0.56,
        full_width_half_max: 0.06
      },
```

```
- assets: {
    - index: {
        type: "text/html",
        title: "HTML index page"
      },
    - thumbnail: {
        title: "Thumbnail image",
        type: "image/jpeg"
      },
    - B1: {
        type: "image/x.geotiff",
      - eo:bands: [
            0
        ],
        title: "Band 1 (coastal)"
      },
    - B2: {
        type: "image/x.geotiff",
      - eo:bands: [
            1
        ],
        title: "Band 2 (blue)"
      },
    - B3: {
        type: "image/x.geotiff",
      - eo:bands: [
            2
        ],
        title: "Band 3 (green)"
      },
```

https://earth-search.aws.element84.com/stac

+ - View source

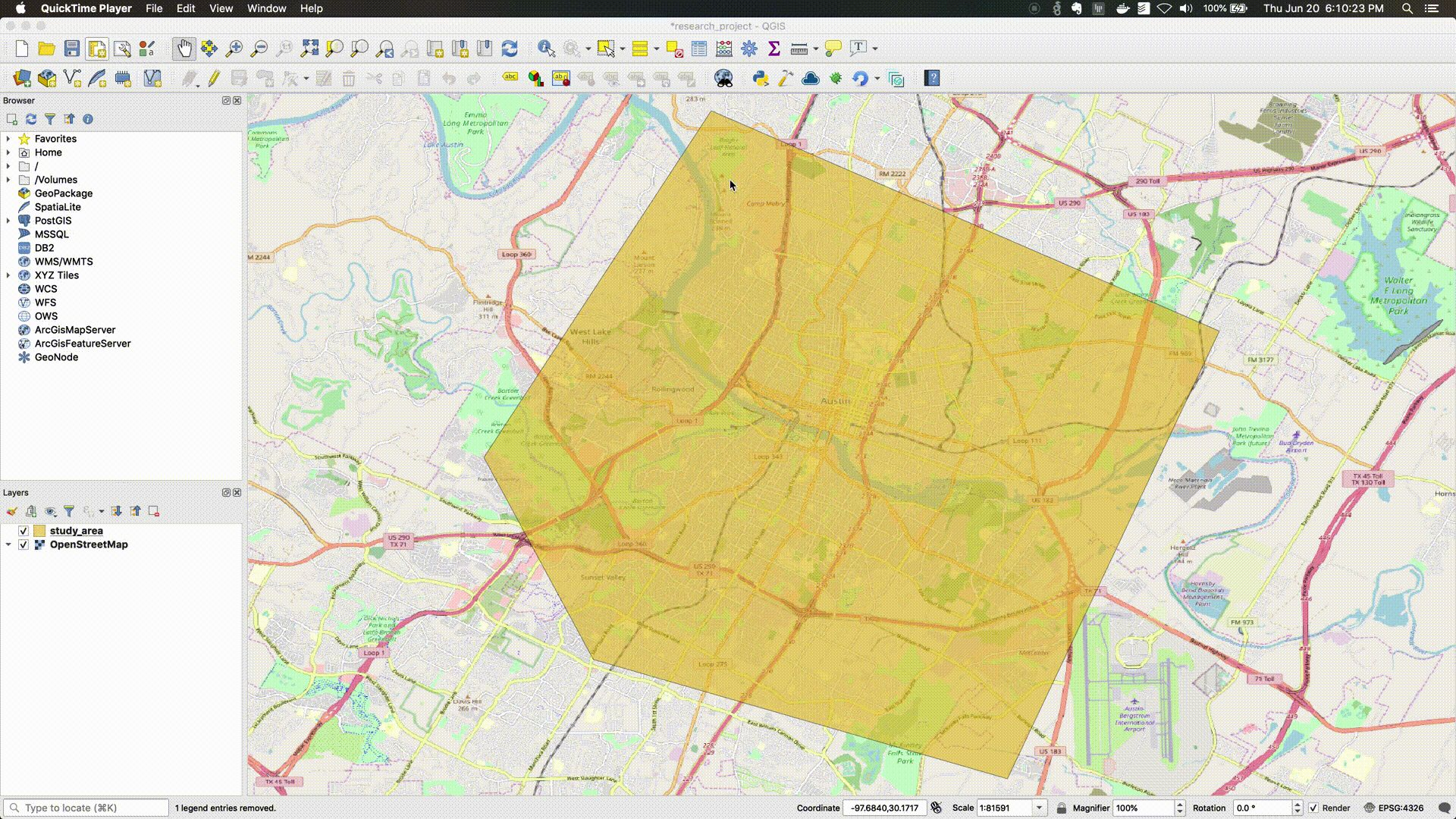

```
{
   stac_version: "0.6.2",
   id: "earth-search",
   title: "Earth on AWS STAC API",
   description: "A STAC API powered by sat-api for Earth on AWS datasets by Element 84",
 - links: [
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-awfi"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan10m"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-mux"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/sentinel-2-l1c"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/landsat-8-l1"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan5m"
       },
     - {
           rel: "self",
           href: "https://earth-search.aws.element84.com/stac"
       }
   ]
}
```


links

STAC Open-Source Ecosystem

Implementation-driven development

- Validators and readers: stac-lint, PySTAC, sat-stac
- APIs: stac-api, staccato, franklin
- Browsers: stac-browser, sat-api-browser, QGIS Plugin
- Clients: sat-search, sat-fetch
- Catalogs: NASA-CMR STAC, E84 Earth-search, SpaceNet, CBERS-4


QuickTime Player File Edit View Window Help
100% Thu Jun 20 6:10:23 PM
*research_project - QGIS
Browser
Favorites
Home
/
/Volumes
GeoPackage
SpatiaLite
PostGIS
MSSQL
DB2
WMS/WMTS
XYZ Tiles
WCS
WFS
OWS
ArcGisMapServer
ArcGisFeatureServer
GeoNode
Layers
study_area
OpenStreetMap
Type to locate (⌘K)
1 legend entries removed.
Coordinate -97.6840,30.1717
Scale 1:81591
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326

SAT-API Browser

BUILD QUERY | RESULTS | CANDIDATES

Date Range

Start Date (required): 2019/07/14 06:33 PM

End Date (required): 2019/07/15 06:33 PM

Area Of Interest (required)

DRAW BBOX ON MAP

Property Filters

Add new filter

Select a property to filter

SUBMIT QUERY



```
$ sat-search search --intersects maine.geojson --found
1674 items found
$ clear
```

Cloud-friendly Data Formats

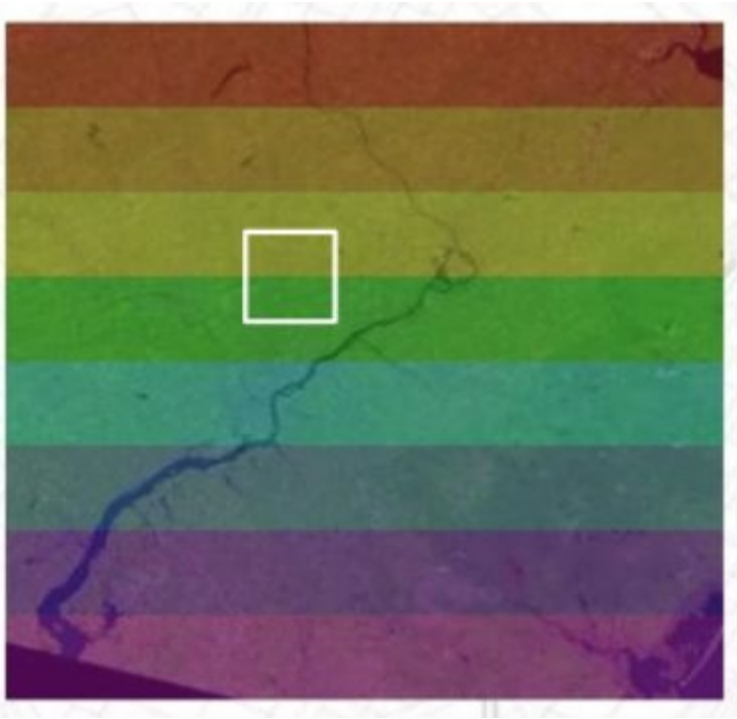
"Hey, let's not download the whole file everytime" - Eugene Cheipesh

- Cloud-optimized GeoTIFF (COG)
- Meta Raster Format (MRF)
- Zarr
- Entwine Point Tiles
- TileDB

Cloud-Optimized GeoTIFF

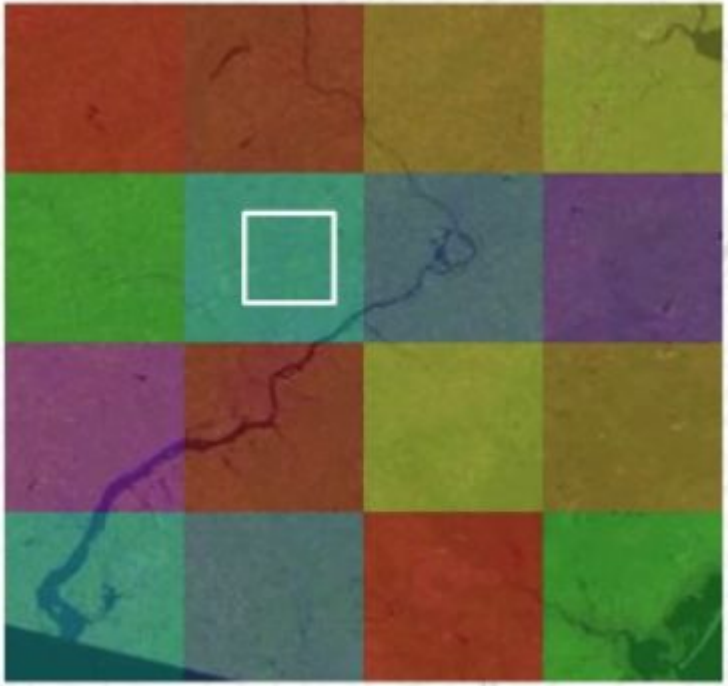
- IFDs (Image File Directory) at beginning of file
  - reduces data seek times
- Uses Tiled Segments
  - decompress and load separately
- Overviews in file
  - efficiently use the data at any resolution

TIFF Striped Segments = BAD



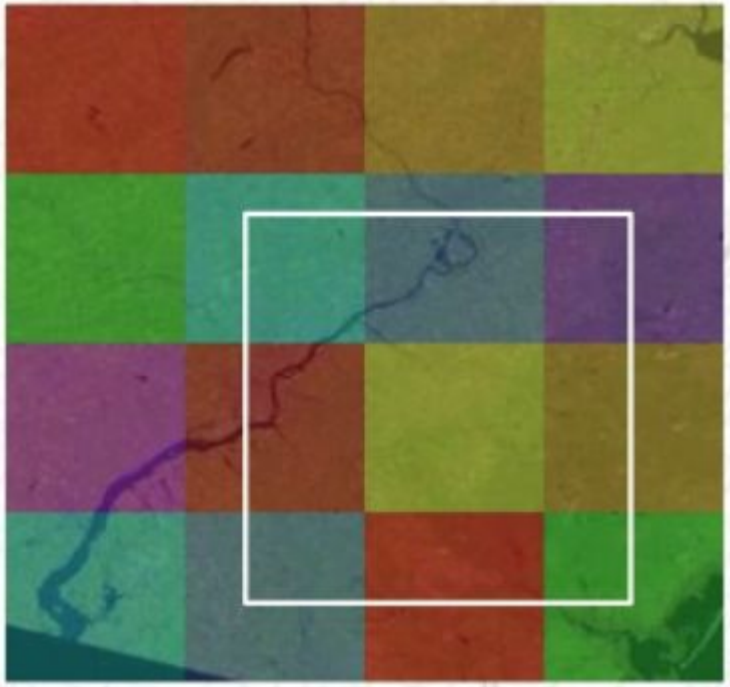


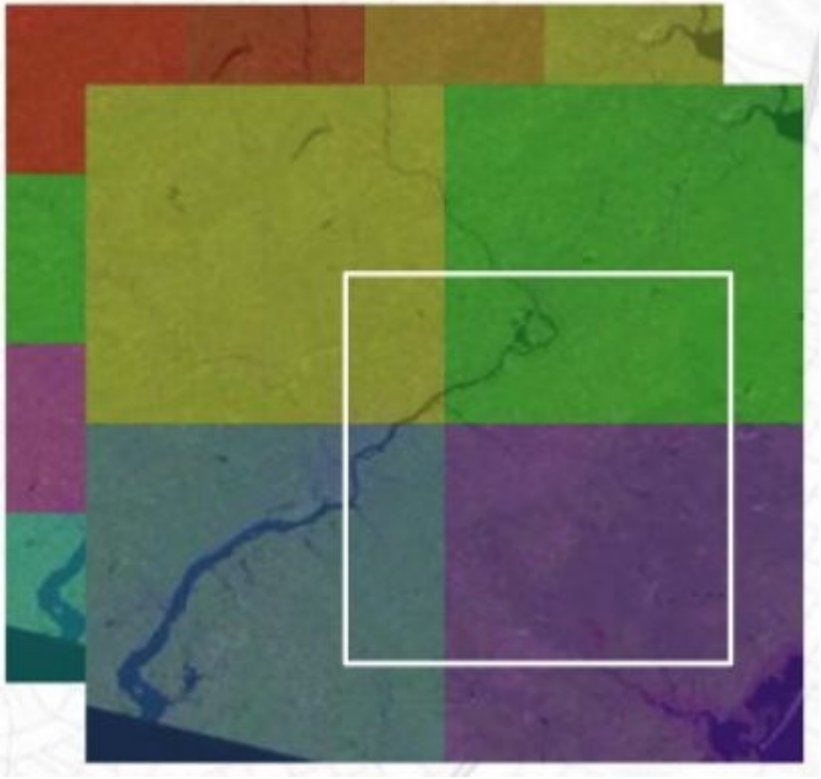
TIFF Tiled Segments = GOOD





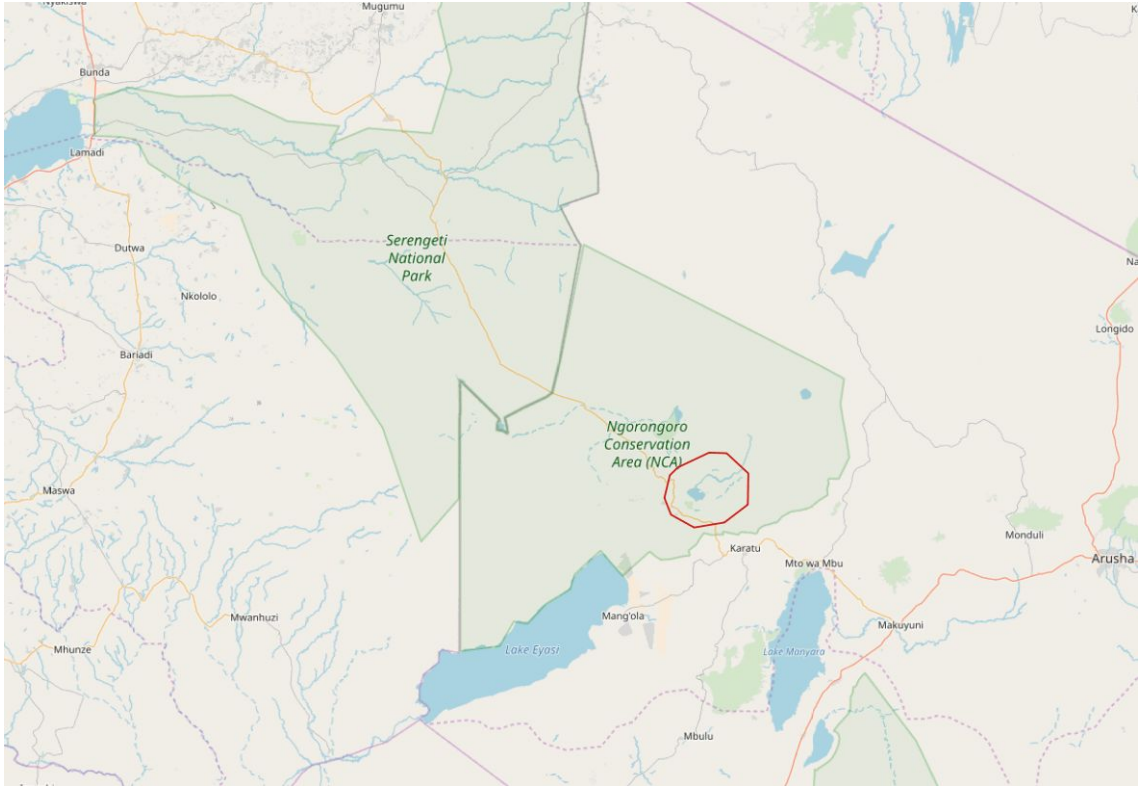
TIFF Tiled Segments = GOOD





STAC
+
COG

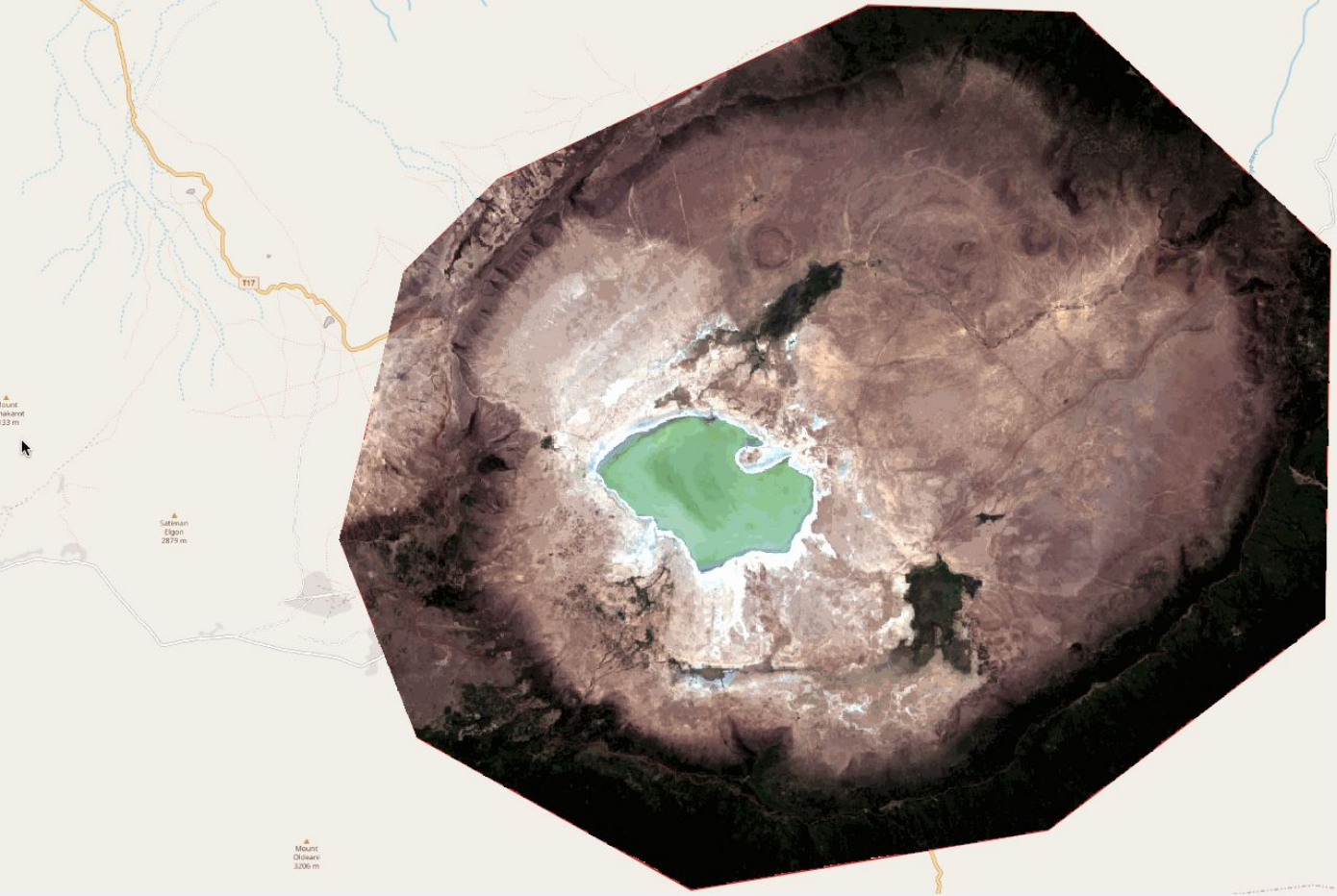
Ngorongoro crater: landsat-8 and sentinel-2



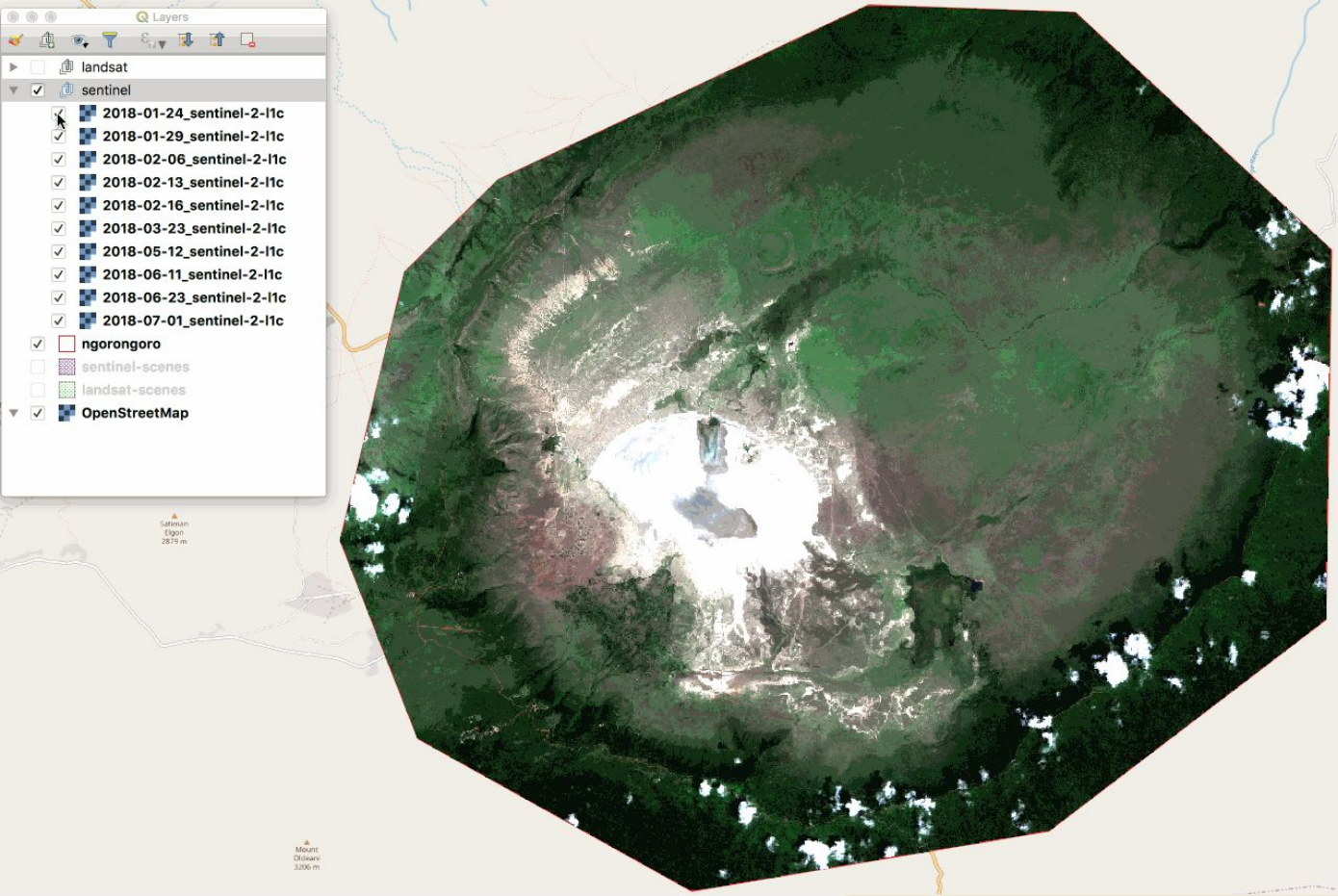
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-12-11  5  LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1  LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0  LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2  LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1  LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0  LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0  LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2  LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0  LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3  LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0  LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2  LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0  LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3  LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```

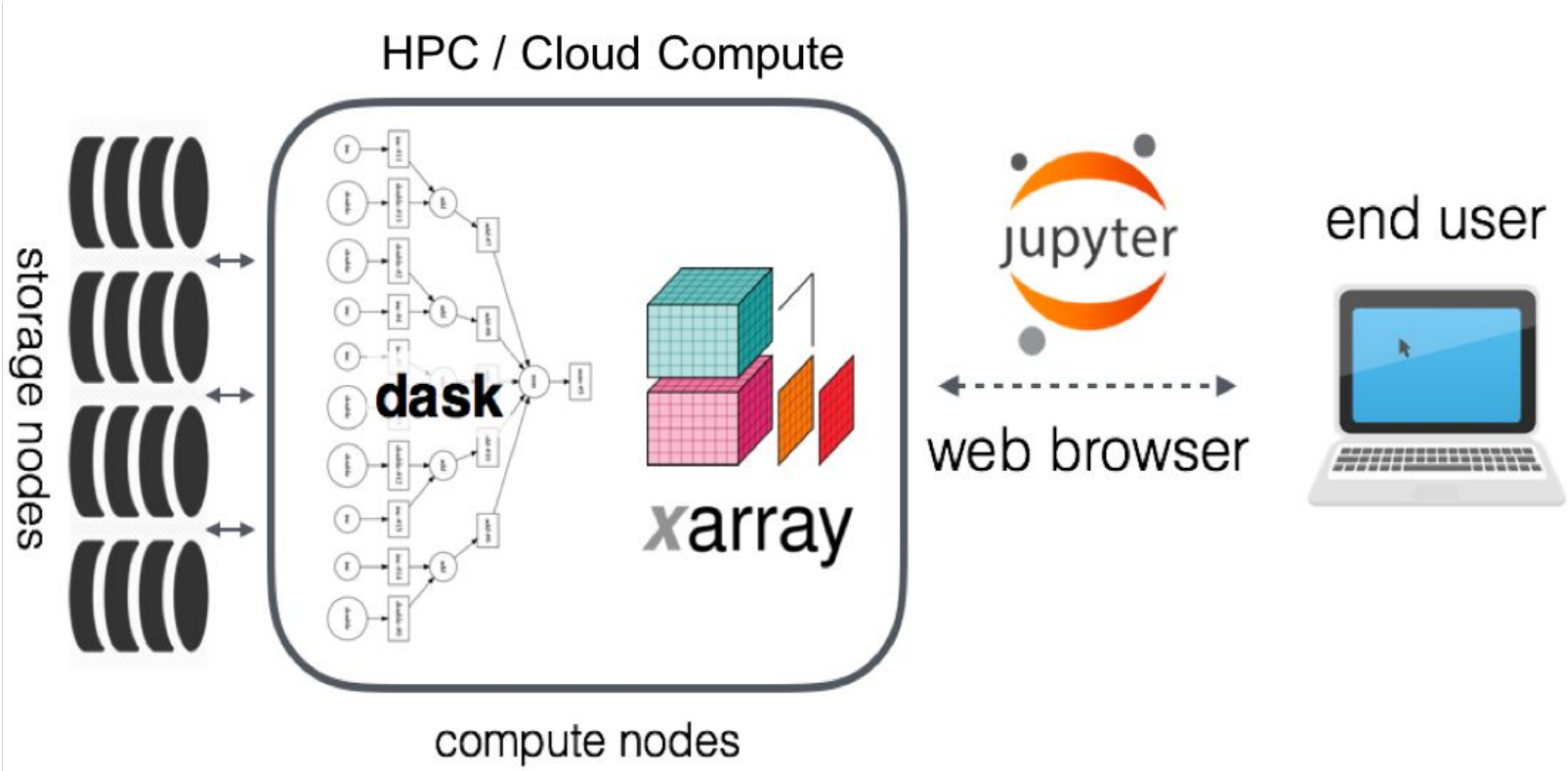



```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```




```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
real 8m6.525s
```

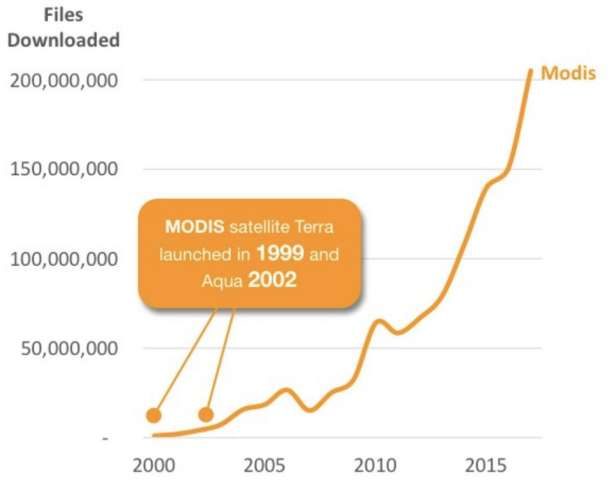
Pangeo

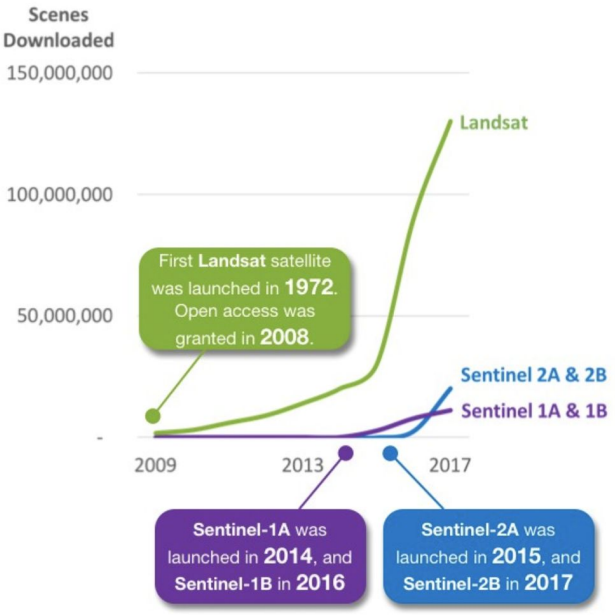

HPC / Cloud Compute
storage nodes
dask
xarray
compute nodes
jupyter
web browser
end user

Diverse Users

Open Satellite Data Downloads

The number of downloads every year from the open data portals has increased exponentially!





SOURCES: MODIS and Landsat data was provided by NASA
Sentinel data was obtained via their annual reports, 2015-2017

*Files downloaded from the MODIS portal often contain multiple scenes

Radiant.Earth
Earth Imagery for Impact

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5.2 OLI Top of Atmosphere Reflectance

Similar to the conversion to radiance, the 16-bit integer values in the Level 1 product can also be converted to TOA reflectance. The following equation is used to convert Level 1 DN values to TOA reflectance:

$$\rho\lambda' = M_\rho * Q_{cal} + A_\rho$$

where:

$\rho\lambda'$ = TOA Planetary Spectral Reflectance, without correction for solar angle. (Unitless)
M_ρ = Reflectance multiplicative scaling factor for the band (REFLECTANCEW_MULT_BAND_n from the metadata).
A_ρ = Reflectance additive scaling factor for the band (REFLECTANCE_ADD_BAND_N from the metadata).
Q_{cal} = Level 1 pixel value in DN

Note that $\rho\lambda'$ is not true TOA Reflectance, because it does not contain a correction for the solar elevation angle. This correction factor is left out of the Level 1 scaling at the users' request; some users are content with the scene-center solar elevation angle in the metadata, while others prefer to calculate their own per-pixel solar elevation angle across the entire scene. Once a solar elevation angle is chosen, the conversion to true TOA Reflectance is as follows:

$$\rho_\lambda = \frac{\rho_\lambda'}{\cos(\theta_{SZ})} = \frac{\rho_\lambda'}{\sin(\theta_{SE})}$$

New users assume data is Analysis Ready


ARD19
August 5-7
2019
USGS Campus
Menlo Park, CA
Geometric and Radiometric Harmonization
Image Quality Metrics, Monitoring and Specification
Machine-learning applications/methods // Time-series Analysis
Fusing satellite imagery with drones, IoT and other non-space sources
Color and Visual Products // Sensor-fusion algorithms
ARD-driven Payload design & System Engineering
Product Development and Application-driven Requirement Gathering
Standards for Data Distribution and Commercialization
Pipelines, Software Engineering and Distributed Computing for ARD

For updates, releases, news:

@GeoSkeptic

Element 84

@element84